Advanced Wireless Communications Laboratory Series

2×2 MIMO-OFDM Detection under Channel Correlation and Estimation Error

GNU Radio and Python simulation laboratory

Reproducible code, recorded measurements, and analysis

WC-LAB-02 2026

Executive summary

This laboratory evaluates zero-forcing (ZF), minimum mean-square-error (MMSE), and ordered MMSE successive-interference-cancellation (MMSE-SIC) detection in a two-stream, 2×2 MIMO-OFDM link. The transmitter uses a 64-point OFDM waveform and a 16-sample cyclic prefix. A six-tap correlated Rayleigh channel is generated in the time domain. Channel estimates are obtained from two orthogonal pilot transmissions; perfect channel knowledge is retained as a reference case.

The recorded study comprises 22 operating points, with 800 independent fading realizations per point. At data $E_s/N_0 = 12$ dB with perfect channel information, increasing the spatial correlation coefficient from 0 to 0.95 increases the median channel condition number from 2.96 to 29.09. In the same comparison, ZF BER rises from 0.02840 to 0.34464; MMSE-SIC BER rises from 0.00757 to 0.12120. At $\rho = 0.6$, a pilot SNR of 20 dB produces a measured channel-estimation NMSE of approximately 0.010 and increases MMSE-SIC BER from 0.01742 (perfect CSI) to 0.02495. All error rates are uncoded.

The experiment isolates two receiver limitations: noise amplification in poorly conditioned channels and residual interference caused by imperfect channel estimates. The results also separate detection performance from implementation burden: direct 2×2 matrix solves require a fixed number of operations, even though their numerical sensitivity and noise enhancement worsen as the channel approaches singularity.

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1 Introduction

Spatial multiplexing transmits independent symbols from multiple antennas on the same OFDM subcarrier. The receiver must separate streams whose coupling is determined by the channel matrix. Spatial correlation can make that matrix poorly conditioned and reduce the reliability of linear detectors [1].

The exercise compares three practical hard-decision receivers. ZF suppresses modeled inter-stream interference without regularizing noise. MMSE balances interference suppression against noise enhancement. Ordered MMSE-SIC uses one detected stream to cancel interference before estimating the second. These methods offer a controlled introduction to the reliability–complexity tradeoff that also motivates more elaborate MIMO receiver implementations [2, 3].

The central question is how detection errors change with the conditioning of $\mathbf{H}[k]$, and how channel-estimation error changes that relationship. No coding, frequency-domain precoding, or adaptive modulation is used; those mechanisms would obscure the detector comparison.

2 System architecture

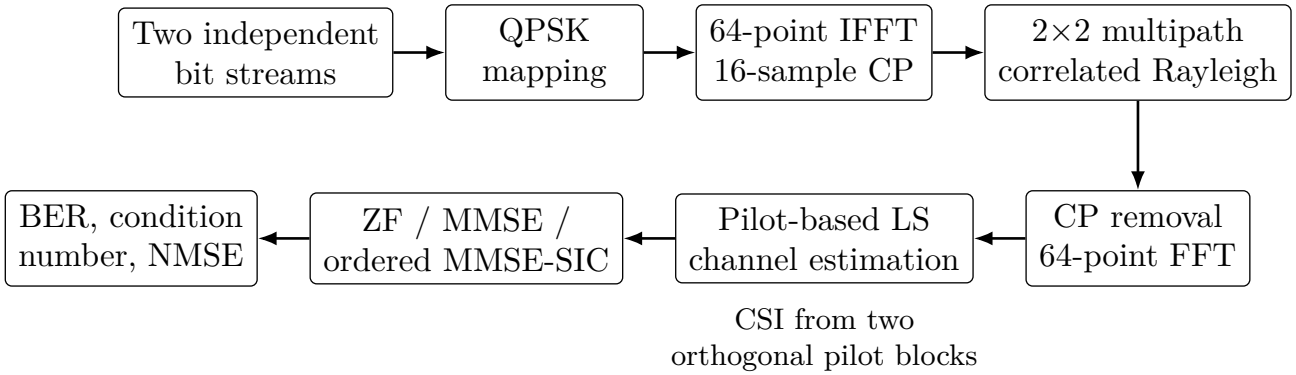


Figure 1: Simulated 2×2 spatial-multiplexing receiver. The same received data and estimated channel are supplied to all three detectors. Pilot observations are formed through the same multipath channel as the data.

2.1 OFDM and channel model

Each transmit antenna uses $N = 64$ FFT bins, of which 52 carry QPSK data. The DC bin and 11 guard bins are unused. The unit-energy constellation is $\mathcal{X} = \{(\pm 1 \pm j)/\sqrt{2}\}$. Two streams therefore carry $2 \times 52 \times 2 = 208$ uncoded bits per data OFDM symbol. The unitary IFFT, cyclic prefix, and linear time-domain convolution are implemented explicitly.

The two-antenna received signal is

$$\mathbf{y}[n] = \sum_{\ell=0}^{5} \mathbf{H}_\ell \mathbf{x}[n-\ell] + \mathbf{w}[n]. \tag{1}$$

The cyclic prefix has 16 samples, longer than the five-sample channel memory. After CP removal and FFT, each occupied tone satisfies

$$\mathbf{y}[k] = \mathbf{H}[k]\mathbf{x}[k] + \mathbf{w}[k], \qquad \mathbf{H}[k] = \sum_{\ell=0}^{5} \mathbf{H}_\ell e^{-j2\pi k\ell/64}. \tag{2}$$

The FFT is unitary, so the complex AWGN variance N_0 is the same in time and frequency. Here $E_s = 1$ per transmitted QPSK symbol, and the stated data SNR is E_s/N_0, not E_b/N_0 or the total received power divided by noise power.

For spatial correlation, the six matrix-valued taps follow

$$\mathbf{H}_\ell = \sqrt{p_\ell}\,\mathbf{R}_r^{1/2}\mathbf{W}_\ell\mathbf{R}_t^{1/2}, \quad p_\ell = \frac{e^{-\ell/1.5}}{\sum_{m=0}^{5} e^{-m/1.5}}, \quad \mathbf{R}_r = \mathbf{R}_t = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}. \tag{3}$$

The entries of $\mathbf{W}_\ell$ are independent $\mathcal{CN}(0,1)$ variables. The parameter ρ lies in $[0,1)$ and denotes the assumed correlation on both transmit and receive arrays. Taps and information bits are independently regenerated for each fading realization. One channel is held fixed over two pilot symbols and one data symbol.

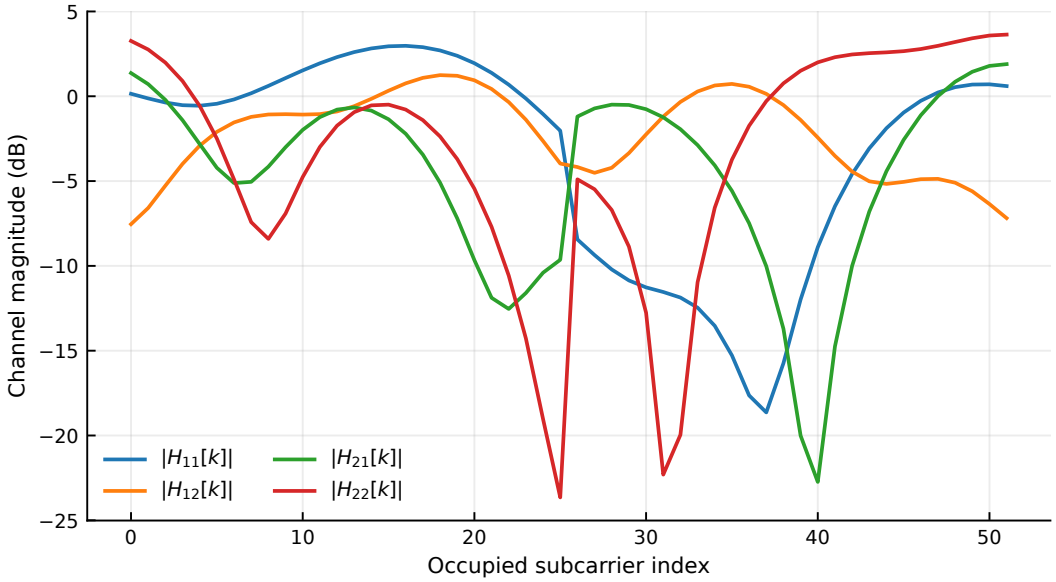


Figure 2: Magnitude of the four frequency-domain channel coefficients for one recorded six-tap realization, $\rho = 0.6$. Frequency selectivity differs across antenna pairs despite the common power-delay profile.

2.2 Channel estimation

Two orthogonal full-band pilot OFDM symbols are transmitted: antenna 1 is active in the first and antenna 2 in the second, with unit pilot symbols on each occupied tone. With $\mathbf{y}_{p,1}[k]$ and $\mathbf{y}_{p,2}[k]$ denoting the two received pilot vectors, the least-squares estimate is

$$\widehat{\mathbf{H}}[k] = \begin{bmatrix} \mathbf{y}_{p,1}[k] & \mathbf{y}_{p,2}[k] \end{bmatrix}. \tag{4}$$

Pilot and data noise are independent. The pilot noise variance is specified by $E_{s,p}/N_{0,p}$, with $E_{s,p} = 1$; it can be changed without altering the data SNR. The estimation error is reported as

$$\mathrm{NMSE} = \frac{\sum_{\text{frames},k} \|\widehat{\mathbf{H}}[k] - \mathbf{H}[k]\|_F^2}{\sum_{\text{frames},k} \|\mathbf{H}[k]\|_F^2}. \tag{5}$$

Perfect CSI is evaluated by setting $\widehat{\mathbf{H}}[k] = \mathbf{H}[k]$. The two dedicated pilot blocks are used to isolate estimator quality; pilot overhead is not optimized or included in a throughput calculation.

3 MIMO detection

For each subcarrier let $\widehat{\mathbf{H}}$ denote the available channel matrix, $\mathbf{y}$ the received vector, and $\widehat{\mathbf{x}}$ the detected symbol vector. The same $\widehat{\mathbf{H}}$ and $\mathbf{y}$ are used by every detector. All reported BER values follow hard QPSK slicing.

3.1 Zero forcing

The ZF estimate solves the least-squares system

$$\widehat{\mathbf{x}}_{\mathrm{ZF}} = (\widehat{\mathbf{H}}^H\widehat{\mathbf{H}})^{-1}\widehat{\mathbf{H}}^H\mathbf{y}. \tag{6}$$

The software uses a linear solve, not an explicitly formed inverse. For full-rank $\mathbf{H}$ with perfect CSI, the output noise covariance is $N_0(\mathbf{H}^H\mathbf{H})^{-1}$. Large inverse singular values therefore imply strong noise enhancement.

3.2 Minimum mean-square error

For independent unit-variance transmitted symbols and AWGN variance N_0,

$$\widehat{\mathbf{x}}_{\mathrm{MMSE}} = (\widehat{\mathbf{H}}^H\widehat{\mathbf{H}} + N_0\mathbf{I}_2)^{-1}\widehat{\mathbf{H}}^H\mathbf{y}. \tag{7}$$

Regularization limits the noise amplification seen in ZF. Under imperfect CSI, this is a plug-in detector: it uses the estimated channel and nominal data-noise variance, without an explicit model for estimation uncertainty.

3.3 Ordered MMSE-SIC

Ordered MMSE-SIC first computes the MMSE estimate and the estimated error covariance $N_0(\widehat{\mathbf{H}}^H\widehat{\mathbf{H}} + N_0\mathbf{I})^{-1}$. The stream with the smaller diagonal error variance is detected first. If p is that stream and q is the other, the receiver makes the hard decision $\widehat{x}_p = \mathcal{Q}([\widehat{\mathbf{x}}_{\mathrm{MMSE}}]_p)$, then forms

$$\mathbf{y}' = \mathbf{y} - \widehat{\mathbf{h}}_p\widehat{x}_p, \qquad \widehat{x}_q = \mathcal{Q}\left(\frac{\widehat{\mathbf{h}}_q^H\mathbf{y}'}{\|\widehat{\mathbf{h}}_q\|_2^2 + N_0}\right). \tag{8}$$

The final decisions are returned in the original stream order. If the first decision is incorrect, its cancellation error can impair the second decision. Ordered SIC and MMSE-based detection are standard receiver concepts; no claim of a new detector is made here [3].

4 Measurement protocol

The channel condition number is computed from the *true* tone-level matrix, not the imperfect channel estimate:

$$\kappa_2(\mathbf{H}[k]) = \frac{\sigma_{\max}(\mathbf{H}[k])}{\sigma_{\min}(\mathbf{H}[k])}. \qquad G_{\mathrm{ZF}}[k] = \frac{1}{2}\,\mathrm{tr}\big[(\mathbf{H}^H[k]\mathbf{H}[k])^{-1}\big]. \tag{9}$$

G_{ZF} is the average noise-enhancement factor over the two streams for perfect CSI. The reported median and 90th percentile summarize values over all occupied tones and frames.

Each detector therefore processes 166,400 bits per point, or 3,660,800 bit decisions over the complete plan. The base seed is 2026. Within each sweep, the underlying information bits, independent fading

Table 1: Reproducible experiment design. All points use 800 independent fading realizations; each contains one data OFDM symbol (208 uncoded bits).

Sweep	Fixed settings and variable	Points
Data SNR	$\rho = 0.3$, perfect CSI; $E_s/N_0 = 0, 4, 8, 12, 16, 20$ dB	6
Spatial correlation	$E_s/N_0 = 12$ dB, perfect CSI; $\rho = 0, 0.3, 0.6, 0.8, 0.95$	5
Pilot quality	$E_s/N_0 = 12$ dB, $\rho = 0.6$; perfect CSI or pilot SNR $= 8, 14, 20, 26$ dB	5
Joint stress	$E_s/N_0 = 12$ dB; $\rho = 0, 0.6, 0.9$ and pilot SNR $= 12, 24$ dB	6
Total	22 points × 800 frames × 208 bits/frame	22

draws, and standardized data-noise samples are shared across operating points; the channel is then transformed by ρ or the noise is scaled by SNR. Pilot noise uses a separate random stream. This pairing reduces extraneous variability in the comparisons. The three detectors always process exactly the same receiver observation at a given point.

BER is the number of erroneous received information bits divided by the total transmitted bits. The CSV includes 95% bootstrap intervals formed by resampling complete OFDM frames 800 times. This treats fading realizations as independent while allowing correlated bit errors among tones of a frame. The intervals describe finite-run uncertainty, not model uncertainty.

5 Recorded results

5.1 Noise sensitivity

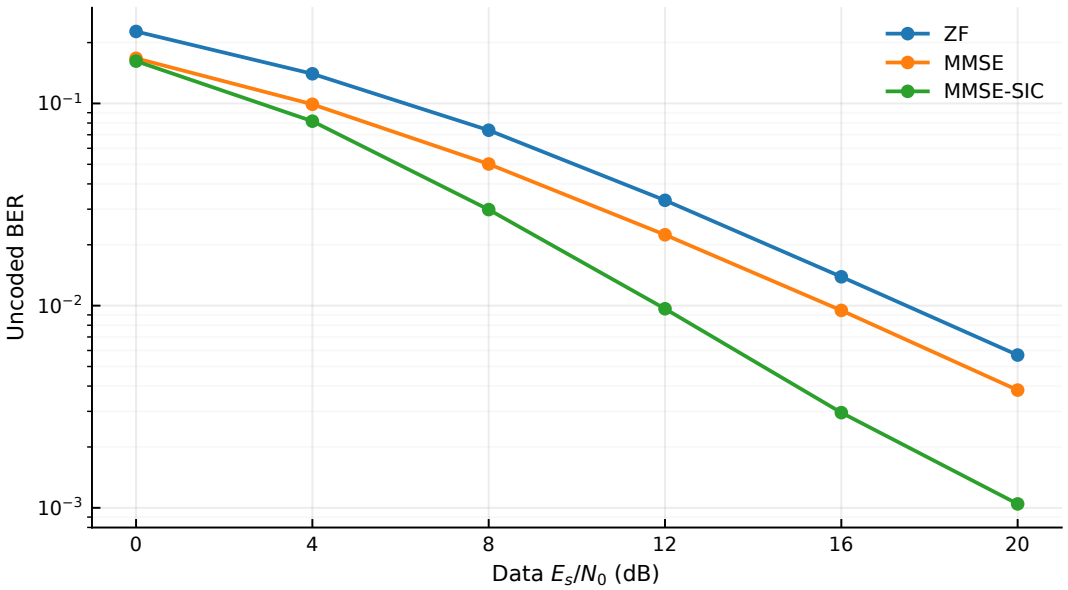


Figure 3: Uncoded BER versus data E_s/N_0 with $\rho = 0.3$ and perfect CSI. All detectors use the same transmitted symbols, channels, and received samples.

All three BER curves improve with SNR. MMSE reduces the noise enhancement of ZF; MMSE-SIC further reduces residual inter-stream interference when its initial hard decision is reliable. At high SNR the differences persist because deep fades and nearly singular channel matrices remain possible.

Table 2: Selected measurements from the data-SNR sweep (perfect CSI, $\rho = 0.3$).

Data SNR (dB)	ZF BER	MMSE BER	MMSE-SIC BER
0	0.22698	0.16722	0.16191
8	0.07371	0.05017	0.02986
12	0.03319	0.02241	0.00965
20	0.00570	0.00382	0.00105

5.2 Spatial correlation and channel conditioning

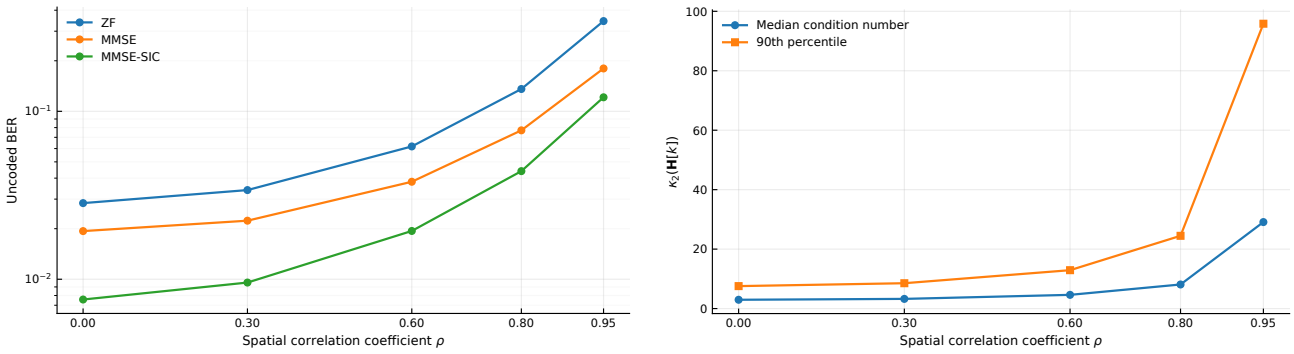


Figure 4: At 12 dB and perfect CSI, increased spatial correlation raises BER (left) and shifts the distribution of channel condition numbers upward (right).

Table 3: Correlation sweep at data $E_s/N_0 = 12$ dB and perfect CSI.

ρ	Median κ_2	Median G_{ZF}	ZF BER	MMSE BER	SIC BER
0.00	2.96	1.65	0.02840	0.01935	0.00757
0.30	3.26	1.93	0.03396	0.02232	0.00956
0.60	4.61	3.66	0.06178	0.03806	0.01939
0.80	8.10	10.80	0.13579	0.07704	0.04403
0.95	29.09	136.24	0.34464	0.17996	0.12120

The median ZF noise-enhancement factor increases from 1.65 to 136.24 as ρ rises from 0 to 0.95. This explains the severe sensitivity of ZF without requiring a change in its algorithmic operation count.

5.3 Pilot-estimation error

Better channel estimates reduce the mismatch in equalization and in the SIC cancellation term. At 20 dB pilot SNR, residual estimation error remains large enough to increase BER for all three detectors relative to perfect CSI.

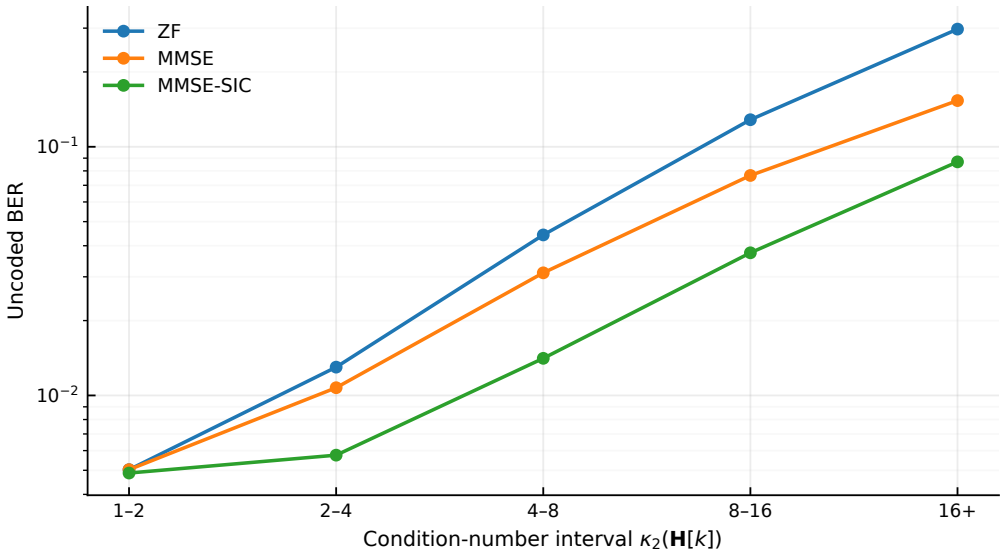


Figure 5: BER grouped by true channel condition number at $\rho = 0.6$, 12 dB, and perfect CSI. Each bin aggregates both streams and all occupied tones. Condition number is a measure of channel geometry, not by itself a complete predictor of BER; channel gain and noise also matter.

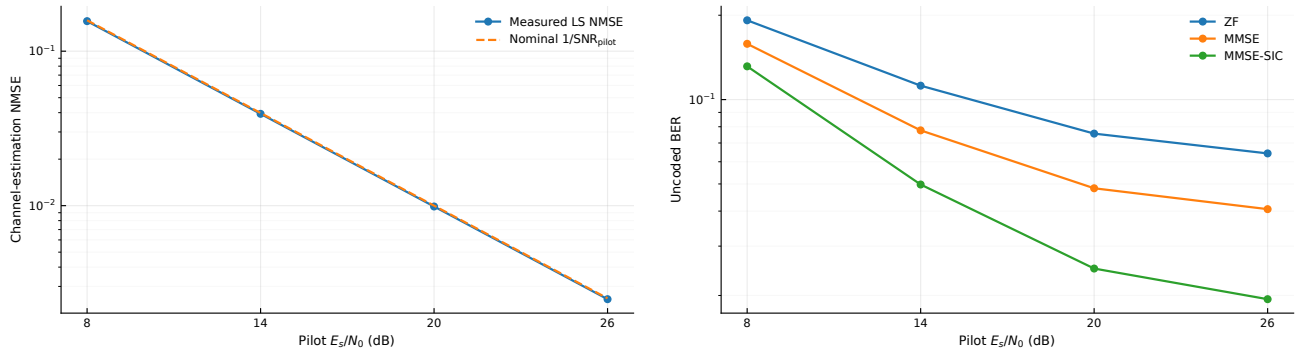


Figure 6: At $\rho = 0.6$ and data SNR 12 dB, LS channel-estimation NMSE (left) and detector BER (right) improve as the independent pilot SNR rises. The $1/\mathrm{SNR}_{\mathrm{pilot}}$ curve is the nominal NMSE for this pilot design.

Table 4: Pilot-quality sweep ($\rho = 0.6$, data SNR 12 dB). Perfect CSI is a separate reference using the same underlying fading and received data samples.

Pilot SNR (dB)	NMSE	ZF BER	MMSE BER	SIC BER
Perfect CSI	0	0.06153	0.03797	0.01742
8	0.156	0.19183	0.15814	0.13141
14	0.039	0.11212	0.07765	0.04977
20	0.010	0.07561	0.04826	0.02495
26	0.002	0.06426	0.04065	0.01940

5.4 Joint correlation and estimation error

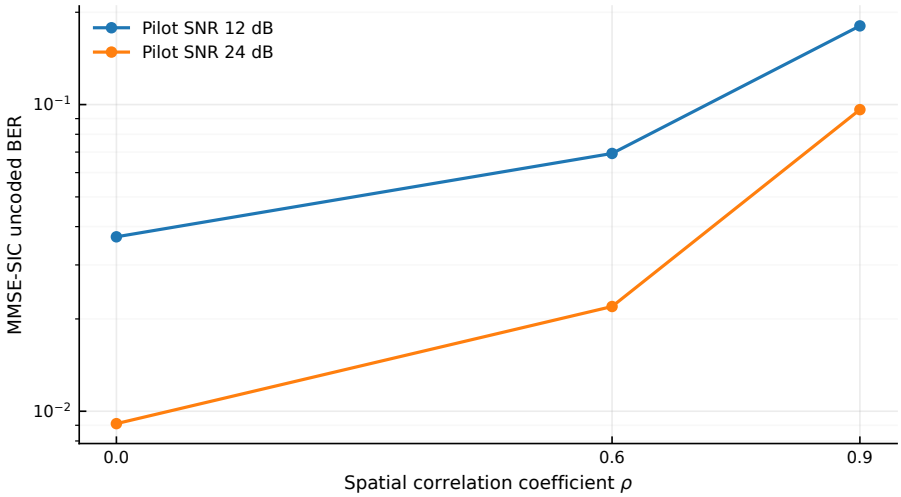


Figure 7: MMSE-SIC BER as correlation and pilot SNR vary together at 12 dB data SNR. Raising pilot quality helps, but cannot undo the loss of spatial separation from a strongly correlated physical channel.

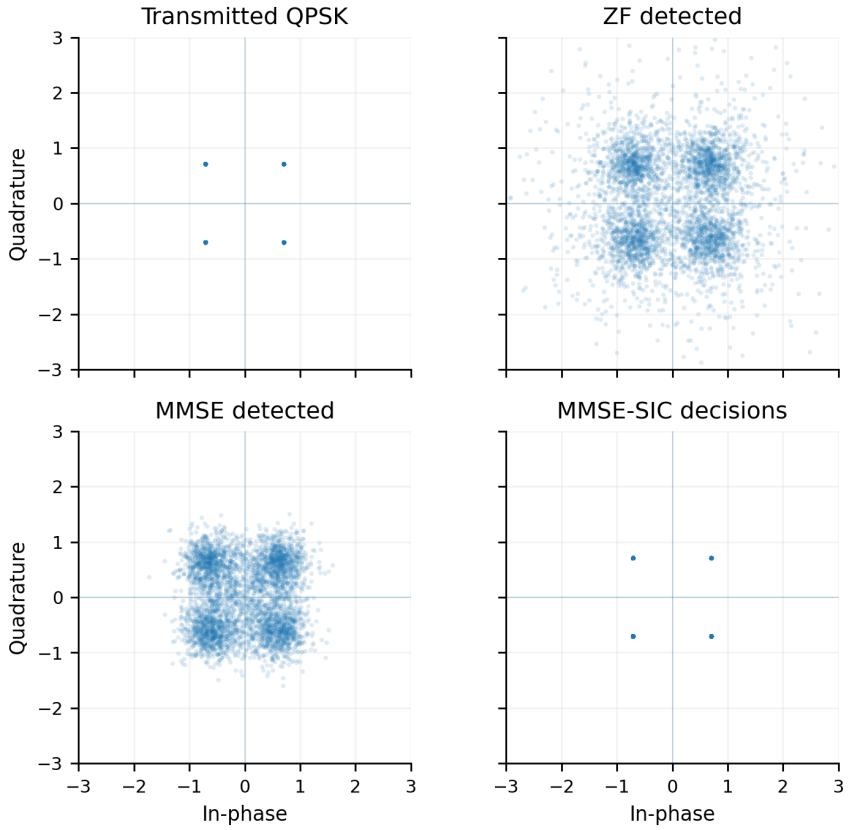


Figure 8: Recorded stream-1 symbols at $\rho = 0.6$, data SNR 12 dB, and pilot SNR 20 dB. ZF and MMSE panels show continuous detector outputs; the SIC panel shows hard QPSK decisions after cancellation.

6 Algorithmic burden and implementation

The comparison uses direct dense solves at each occupied tone. The dominant operations are summarized below; the MMSE factorization and its solution are reused in the SIC branch.

Table 5: Per-tone algorithmic structure. Counts describe the receiver method, not measured CPU cycles, chip area, or hardware energy.

Detector	Required operations
ZF	Form $\widehat{\mathbf{H}}^H\widehat{\mathbf{H}}$ and solve one 2×2 system; hard slice
MMSE	Regularized 2×2 solve; hard slice
Ordered MMSE-SIC	Regularized 2×2 solve with additional right-hand sides for ordering; one hard decision, cancellation, scalar projection, final hard decision

These operation types do not depend on the numerical condition number. Poor conditioning increases noise amplification for ZF and can increase finite-precision sensitivity; it does not make a 2×2 direct solve perform more iterations. MMSE regularization changes the spectrum of the system matrix. SIC incurs additional decisions and cancellation, and can propagate an incorrect first decision. The distinction between algorithm complexity and hardware realization is also central to the MIMO detection literature [2].

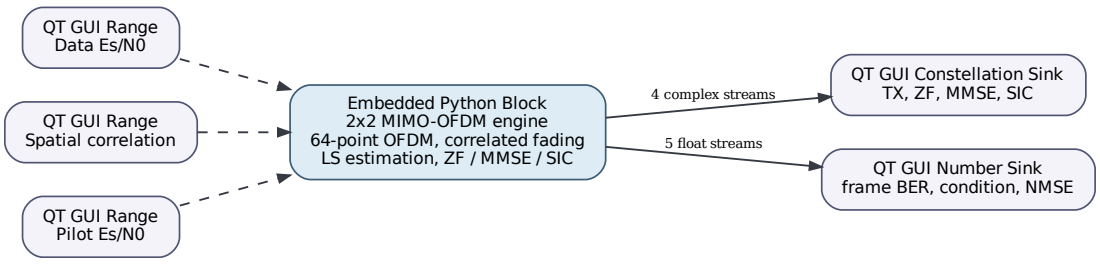


Figure 9: Signal connections in the GNU Radio Companion flowgraph. The Embedded Python Block implements the full two-stream transmitter, fading channel, pilot estimation, and three detectors. The QT GUI sinks present symbol constellations and per-frame diagnostic values.

The Python simulation generated all 22 operating points and recorded aggregate detector measurements in two CSV files. The GNU Radio flowgraph displays signal constellations and per-frame detector statistics.

7 Scope and limitations

The simulation is an uncoded, perfectly synchronized, 2×2 link. The fading matrix is constant over the two pilot blocks and one data block, with independent realizations between frames. Both arrays use the same single-parameter correlation model; no array geometry, Doppler, phase noise, frequency offset, pilot interpolation, or channel-estimator tracking is modeled. The pilot allocation is deliberately simple and does not constitute a throughput-efficient training design. MMSE uses nominal AWGN variance and does not compensate explicitly for channel-estimation uncertainty.

The measurements compare the three detector algorithms under the stated channel and pilot-estimation conditions.

8 Conclusion

The experiment confirms that spatial correlation and CSI error affect 2×2 spatial-multiplexing reception in different, complementary ways. Correlation reduces channel separability and amplifies ZF output noise; noisy pilots bias the detector and the SIC cancellation. MMSE improves reliability over ZF in every reported configuration, and ordered MMSE-SIC produces the lowest measured BER among the three methods at all 22 points. Its additional cancellation work is fixed for a two-stream receiver and must be assessed against the achieved error-rate improvement. This establishes a transparent baseline for subsequent work on detector quantization, channel estimation, and implementation efficiency.

References

[1] D. W. M. Guerra, R. M. Fukuda, R. T. Kobayashi, and T. Abrão, "Efficient detectors for MIMO-OFDM systems under spatial correlation antenna arrays," *ETRI Journal*, vol. 40, no. 5, pp. 570–581, 2018. `https://doi.org/10.4218/etrij.2018-0005`.

[2] C. Studer, A. Burg, and H. Bölcskei, "Soft-output sphere decoding: Algorithms and VLSI implementation," *IEEE Journal on Selected Areas in Communications*, vol. 26, no. 2, pp. 290–300, Feb. 2008. `https://doi.org/10.1109/JSAC.2008.080206`.

[3] D. Wübben, R. Böhnke, V. Kühn, and K.-D. Kammeyer, "Near-maximum-likelihood detection of MIMO systems using MMSE-based lattice-reduction," in *Proc. IEEE International Conference on Communications*, Paris, France, June 2004. `https://www.ant.uni-bremen.de/en/publications/1996/`.